



NATIONAL OCCUPATIONAL STANDARD FOR A LIVESTOCK ARTIFICIAL INSEMINATOR

**NOS.LAI.01
FIRST EDITION**

APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 19th May 2022.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised after every **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Agriculture National Occupational Standards Development Team, upon which the following organisations were represented:

1. Agricultural Institution of Zambia (AIZ)
2. Aquaculture Development Association of Zambia
3. CropLife Zambia
4. Ministry of Fisheries and Livestock (Department of Fisheries)
5. Golden Valley Agricultural Research Trust (GART)
6. Katete College of Agricultural Marketing
7. Ministry of Agriculture (Department of Agriculture)
8. Mulungushi University (MU)
9. Natural Resources Development College (NRDC)
10. University of Zambia (UNZA)
11. Veterinary Council of Zambia
12. Zambia Agriculture Research Institute (ZARI)
13. Zambia National Farmers Union (ZNFU)
14. Zambia Seed Trade Association (ZASTA)
15. Zambia Qualifications Authority (ZAQA)

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to “**provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing**”.

Among other functions, ZAQA is responsible for “**determining national standards for any occupation**”, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Agriculture National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes, in various sectors where the occupation exists.

JUSTIFICATION

Artificial Inseminators are necessary to plan and carry out artificial insemination in livestock. This task ensures that the best genetics can be utilised in each situation, and that Dairies, can breed the required traits into their new heifers.

An Artificial Inseminator, has the capability to plan and oversee an entire artificial insemination program, including hormonal synchronisation (where necessary), heat detection programs, and the artificial insemination process.

ACRONYMS AND ABBREVIATIONS

AI	Artificial Insemination
CS	Core Skill
LAI	Livestock Artificial Inseminator
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Knowledge
PC	Performance Criteria
PS	Professional Skill
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
TK	Technical Knowledge
VCZ	Veterinary Council of Zambia
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

This overview gives an outline of the key requirements and information regarding the Artificial Inseminator, their job purpose, and the level of qualification required.

NOS Code	NOS.LAI.01
Occupation	Artificial Inseminator
Job Title	Livestock Artificial Inseminator
Job Description	The role of the Livestock Artificial Inseminator is to plan and implement an artificial insemination program, including synchronization of heat. The role also includes a working knowledge of how to store, transport, thaw and prepare semen for insemination, and to offer reproductive consultation to breeders and farmers of livestock.
Job Purpose	To ensure effective and efficient livestock insemination activities at the farm/facility in order to optimize reproductive efficiency in animals
ZQF Level	6
Sector	Agriculture
Sub sector	Livestock
Other Economic Sector(s) in which the Occupation is Practiced	Education Sector (Academic research units), Agricultural Sector (Semen sellers, Semen Collection Centres, Cattle Embryo Transfers Units, Dairy Farms, Beef Farms, Breeding Centres)
Other Similar Jobs that can be performed by an Artificial Insemination Technologist	Advisors and Practical instructors
Minimum Educational Job Entry Qualification(s)	Diploma in Animal Health
Practicing License Requirements (if any)	Veterinary Council of Zambia
Training/RPL	Specialist training, and experience in Artificial Insemination (either within the diploma, or as an extra module).
Minimum Job Entry Age	20 years
Prior Experience	Six months practicing AI, under supervision, in the livestock industry, either within the diploma or as extra experience.
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competencies that an Artificial Inseminator must possess to be successful in his/her job role. It is applicable to Livestock Artificial Inseminators working in public or private organisations or self-employed.

The Artificial Inseminator must have an understanding of the internal physiology of the cow, and is able to plan and execute a heat synchronization program, as well as a heat detection program, in the run-up to an artificial insemination program. They need to know how to do artificial insemination across a wide range of species.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Organisational skills, to arrange trainings
- Mathematical skills, for calculation of drug dosages
- Interpersonal skills
- Gentleness
- Attention to detail with good record-keeping skills
- Ability to communicate effectively and clearly
- Self-motivated and team worker
- Ability to plan and prioritize,
- Quality consciousness and time conscious
- Occupational health and safety oriented

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into seven (07) Units representing the tasks that a job holder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit is about the Artificial Inseminator training others on the procedure]

Unit No.	01
Unit Title	Conducting of Trainings in Livestock Artificial Insemination
Description	This unit is about the knowledge and skills required in order for a technologist to conduct trainings of Livestock Artificial Insemination.
Scope	This unit covers the following: • Health & safety requirements • Advantages and Disadvantages of AI compared to natural breeding. • Anatomy of the Reproductive tract in each domestic species • Physiology of each specie's oestrous cycle, and the importance of only doing AI in healthy, well-nourished animals (knowledge of basic nutrition) • The Artificial insemination process, including handling of semen, and the liquid nitrogen tank, heat detection, and heat detection aides • The Synchronisation of a herd of animals, and how to detect heat • Preparation of training materials and presentations for the training process.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & safety regulations	To be competent, the individual must be able to: PC1. Read, interpret and implement Health and Safety Standards PC 2. Assess risks and possible safety hazards of all aspects of operations
Advantages and Disadvantages of AI compared to normal breeding	To be competent, the individual must be able to: PC3. Articulate the advantages and disadvantages of Artificial Insemination and determine the best situations for the use of AI.
Anatomy of the Reproductive tract in each domestic species	To be competent, the individual must be able to: PC4. Explain and annotate each species reproductive tract and explain the potential barriers to the introduction of the AI pipette
Physiology of each specie's oestrous cycle, and the importance of only doing AI in healthy, well-nourished animals (knowledge of basic nutrition)	To be competent, the individual must be able to: PC5. Explain the physiology of each species oestrous cycle and the timing of insemination. PC6. Explain the importance of only doing AI in healthy, well-nourished animals.

Artificial insemination process, including handling of semen, and the liquid nitrogen tank, heat detection, and heat detection aides	To be competent, the individual must be able to: PC7. Explain and demonstrate the safe handling of semen straws or semen sachets. PC8. Explain and demonstrate the safe handling of liquid nitrogen. PC9. Explain the equipment required for an artificial insemination program in each species PC10. Explain and understand how to evaluate the success of an artificial insemination program.
Synchronisation of a herd of animals, and how to detect heat	To be competent, the individual must be able to: PC11. Explain the principles of heat detection and the usage of heat detection aides. PC12. Explain the principles of synchronisation and the usage of drugs to manipulate breeding
Preparation of training materials and presentations for the training process	To be competent, the individual must be able to: PC13. Prepare materials required for the theoretical and practical aspects of the training course. PC14. Prepare the presentations required for the training course.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Organisation/Facility Quality, health and safety policies OK2. Organisation/Facility regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Safety and health risk assessment for the training of students. TK2. Hormones and when to use them. TK3. Administration of hormones to control the breeding cycle. TK4. The best timing for insemination. TK5. The equipment and how to use it. TK6. Monitoring of the success of an AI program TK7. Assess whether an animal can be considered to be healthy and well-nourished enough for AI. TK8. Operate a computer and power-point projector.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory requirements for health & safety. RK2. The requirements to be registered by the relevant regulatory body.
Skills (S)	
	Writing Skills

A. Core Skills/ Generic Skills	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, health and safety instructions, memos, and other company documents, including pamphlets on hormonal drugs being used, and how to calculate dosages correctly. CS4. Read from different sources- books, screens in machines, and signage. CS5. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS6. Express statements or information clearly so that others can hear and understand. CS7. Participate in and understand the main points of simple discussions. CS8. Respond appropriately to any queries.
B. Professional Skills	Working, practical knowledge of the AI process
	The individual on the job must be able to: PS1. Competently demonstrate the whole AI process from start to finish
	Plan and organise
	The individual on the job must be able to: PS2. Plan and organise an artificial insemination training course, including organisation of materials for practical sessions, including loose uteruses, and cull cows for practicing the process. PS3. Work constructively and collaboratively with others.
	Explain and teach
	The individual on the job must be able to: PS4. Follow the industry code of conduct. PS5. Manage relationships with clients and farmers with the intent of satisfying its requirements for service delivery.

UNIT 2 [This unit is about the requirements for planning and implementation of a heat synchronisation program]

Unit No.	02
Unit Title	Planning and Implementation of a heat synchronisation program
Description	This unit is about the knowledge and skills required in order for a technologist to use hormonal drugs in order to manipulate the timing of the heat cycles in a group of cows, in order to better facilitate the AI process. .
Scope	This unit covers the following: • Health and Safety, with regard to usage of medicines • The Synchronisation Process (including hormonal medicines, the different types of synchronisation programs, the best timing of artificial insemination for each programme, the equipment required for the synchronisation process, and monitoring of the success of a synchronisation program)
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health and Safety, with regard to usage of medicines	To be competent, the individual must be able to: PC1. Read, interpret and implement Health and Safety Standards with regard to the usage of medicines. PC 2. Assess risks and possible safety hazards of all aspects of operations.
Synchronisation Process	To be competent, the individual must be able to: PC3. Explain the physiology of the oestrous cycle of each species, and physically detect the stage in which an animal is in. PC4. Identify the different types of hormonal remedies and the effect they have on the oestrous cycle. PC5. Use hormonal remedies and calculate the correct dosages. PC6. Explain the different programs which are available. PC7. Plan and implement a synchronisation program, and ensure that all equipment and consumables are procured. PC8. Use standard mathematical formulae to evaluate the success of a synchronisation program.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/orga nisation)	The individual on the job must demonstrate knowledge and understanding of: OK1. The farmers objectives with regards to the AI program. OK2. Knowledge of standard operating procedures of the organisation.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of:

	TK1. Different species oestrous cycle and how to detect what stage each animal is in. TK2. The hormonal remedies available and when to best use them. TK3. The different programs available. TK4. Evaluating the success of a program.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory requirements for health & safety
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical and honest manner to maintain integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to:

	PS1. Follow organisation rule-based decision-making process. PS2. Take decisions with a systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with the intent of satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognise problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 3 [This unit is about requirements for planning and implementation of a heat detection program]

Unit No.	03
Unit Title	Planning and Implementing a heat detection program
Description	This unit is about the knowledge required in order for a technologist to plan a program to accurately detect animals that are in heat so that they can be artificially inseminated.
Scope	This unit covers the following: • Detection of heat in each species • Planning Heat Detection Programs.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Detection of Heat in each species	To be competent, the individual must be able to: PC1. Plan a program to detect heat in a group of animals. PC2. Determine the roles of each person in the project team.
Planning Heat Detection Programs	To be competent, the individual must be able to: PC3. Plan a heat detection program for a particular species, and physically detect heat in an individual animal. PC4. Determine the frequency of timing of heat detection. PC5. Determine the aides needed to help with heat detection.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company Quality, health, and safety policies OK2. Company regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. The signs each species shows when in heat. TK2. Times per day (frequency) heat detection should be done for each species. TK3. The best times of day to do heat detection. TK4. Knowledge of how long to monitor each species for heat detection. TK5. Using the different heat detection aides. TK6. When to present the animal for AI following detection of heat.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health & safety
	Writing Skills

A. Core Skills/ Generic Skills	The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, and other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature, and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully, and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical, and honest manner to maintain the integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decisions with a systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to:

	PS7 Recognise problems, and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 4 [This unit is about planning and implementing an artificial insemination program]

Unit No.	04
Unit Title	Planning and Implementing an artificial insemination program
Description	This unit is about the knowledge and skills required in order for a technologist to plan the artificial insemination program, and carry it out
Scope	This unit covers the following: • Health & safety regulations • Planning an Artificial Insemination program
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & safety regulations	To be competent, the individual must be able to: - PC1. Read, interpret and implement Health and Safety Standards, especially with respect to the usage of liquid nitrogen and handling of the cow. - PC2. Assess risks and possible safety hazards of all aspects of operations
Planning an Artificial Insemination Program	To be competent, the individual must be able to: - PC3. Plan an artificial insemination program for a group of animals. - PC4. Select the correct sire for each breeding, by looking at different genetic traits. - PC5. Prepare all equipment and consumables needed for the AI program. - PC6. Correctly store the semen for each species (liquid nitrogen for livestock, and fridge for sheep/goats/pigs). - PC7. Carry out the artificial insemination process.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company Quality, health, and safety policies OK2. Company regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Accurate record-keeping. TK2. Good health and nutrition. TK3. Planning an artificial insemination program. TK4. Selection of the best semen to use. TK5. Storage of semen for each species. TK6. Carrying out the practical AI process from start to finish, for each species. TK7. Using mathematical formulae to objectively analyse the success rate of an AI program, and to compare results of different inseminators.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements for health & safety
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully and professionally with a range of stakeholders using appropriate resources and techniques

	CS11. Communicate in a culturally sensitive, ethical and honest manner to maintain integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decisions with a systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with the intent of satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognise problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations, and data to select the course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 5 [This unit is about the heat detection process]

Unit No.	05
Unit Title	The heat detection process.
Description	This unit is about the knowledge and skills required in order for a technologist to accurately detect animals that are in heat so that they can be artificially inseminated.
Scope	This unit covers the following: • Health & safety regulations • Heat Detection
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & safety regulations	To be competent, the individual must be able to: PC1. Follow Health and Safety Standards
Heat Detection	To be competent, the individual must be able to: PC2. Monitor a herd of animals, and determine which ones are in heat. PC3. Analyse any aides being used to help with heat detection. PC4. Know when to present the animal for artificial insemination, after detecting it is in heat.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: - OK1. Company Quality, health, and safety policies with respect to handling of cows. - OK2. Company regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. The primary signs an animal shows when in heat, as well as a knowledge of secondary signs and their significance. - TK2. Times per day (frequency) heat detection should be done. - TK3. The best times of day to do heat detection. - TK4. Knowledge of how long to monitor a group of animals for heat detection. - TK5. Using the different heat detection aides. - TK6. When to present the animal for AI following detection of heat.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Government regulatory agency requirements for health & safety

A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, and other company documents. CS5. Read from different sources- books, screens in machines, and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully, and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical, and honest manner to maintain the integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decisions with a systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making

	The individual on the job must be able to: PS7 Recognise problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations, and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 6 [This unit is about carrying out artificial insemination]

Unit No.	06
Unit Title	The artificial insemination Process
Description	This unit is about the knowledge and skills required in order for a technologist to carry out the AI process.
Scope	This unit covers the following: • Health & safety regulations • Artificial Insemination in Cattle • Artificial Insemination Process in sheep, goats and pigs
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & safety regulations	To be competent, the individual must be able to: PC1.Implement Health and Safety Standards
Artificial Insemination Process in Cattle	To be competent, the individual must be able to: PC2. Competently and safely remove the correct semen straw from the liquid nitrogen flask. PC3. Introduce the straw into a pre-prepared warm water bath, at the required temperature, within the required time-frame. PC4. Thaw the straw for the required amount of time. PC5. Remove the straw, confirm the identification of the semen, and then load the straw correctly into the AI pipette. PC6. Cut the straw at the correct angle, and insert the pipette into a sterile sheath. PC7. Introduce the sheath into a plastic sleeve (optional for extra hygiene) and then wrap the pipette correctly, and store it to keep it at the correct temperature. PC8.Approach the restrained cow, and confirm its identification.

	PC9. Introduce the non-AI hand into the rectum, and correctly secure the cervix. PC10. Hygienically introduce the AI pipette, breaking the sleeve as necessary, and pass the pipette through the cervix. PC11. Deposit the semen in the required place (less than 0.5 cm into the uterus). PC12. Remove the pipette, and gently massage the uterus. PC13. Inspect the pipette to ensure that the process has been completed correctly, and that there are no anomalies. PC14. Write down the semen straw details, before discarding it. PC15. Keep the necessary records to record the process. PC16. Dispose of waste materials correctly.
Artificial Insemination Process in sheep, goats and pigs	To be competent, the individual must be able to: PC17. Remove the semen from the fridge and thaw at room temperature. PC18. Competently prepare the correct catheter/pipette/rod for the animal. PC19. Restrain the animal, make sure the vulva is cleaned, and introduce the catheter/rod. PC20. Lock the rod into the cervix (pigs), attach the semen sachet, and inseminate the animal. PC21. Remove the empty sachet, but leave the rod in place for 1-2 more minutes. PC22. Inspect the rod/pipette to ensure there are no anomalies that need to be recorded.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. The organisations quality, health, and safety policies OK2. The organisations regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Handling semen and liquid nitrogen TK2. Thawing the semen straws/sachets and load the AI pipette/rod. TK3. Keep the thawed semen viable until insemination is carried out. TK4. Inseminate the animal. TK5. Keep the required records.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory requirements for health & safety

A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, and other company documents. CS5. Read from different sources- books, screens in machines, and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical and honest manner to maintain integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decisions with a systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with the intent of satisfying their requirements for service delivery.
	Problem Solving and Decision Making

	The individual on the job must be able to: PS7. Recognise problems and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations, and data to select the course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 7 [This unit is about Health, Safety, and Environmental Management].

Unit No.	07
Unit Title	Safety, health, hygiene, and environmental management
Description	This unit is about maintaining safety, health and environmental protection for the individual and the plant
Scope	This unit covers the following: • Health & safety policies for the workplace • General hygiene policies with respect to the artificial insemination process • Environmental protection, and waste management.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & safety policies for the workplace	To be competent, the individual must be able to: PC1. Implement organisational safety and health policies and regulations. PC2. Prevent risks and possible safety hazards in all aspects of operations
General hygiene policies with respect to the artificial insemination process	To be competent, the individual must be able to; PC3. Follow the organisation's hygiene policies PC4. Follow the organisations cleaning schedules.
Environmental protection, and waste management	To be competent, the individual must be able to: PC5. Implement the environmental policies for the organisation, especially with regard to waste management. PC6. Implement environmental standard operating procedures and policies of the organisation

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company Quality, health and safety policies OK2. Company environmental policies OK3. Company regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Safety and health risk assessment TK2. Environmental Risk assessment TK3. Hygiene Policies and Procedures TK4. Health and Safety Policies and Procedures
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Regulatory requirements for operations of a health & safety committee.
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions. CS2. Write complex nutritional reports and recommendations for specific situations.
	Reading Skills
	The individual on the job must be able to: CS3. Read and interpret internal/external documents. CS4. Read and understand manuals, health and safety instructions, memos, and other company documents. CS5. Read from different sources- books, screens in machines and signage. CS6. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Express statements or information clearly, so that others can hear and understand. CS8. Participate in and understand the main points of discussions. CS9. Respond appropriately to any queries. CS10. Communicate scientifically, truthfully and professionally with a range of stakeholders using appropriate resources and techniques CS11. Communicate in a culturally sensitive, ethical and honest manner to maintain integrity of the animal nutrition profession and the individual. CS12. Translate scientific evidence and concepts to simple and safe messages, following legislative requirements

	for safe practice that can be understood by a variety of consumer and non-science groups. CS13. Recognise own limitations to relevant nutrition profession scope of practice and values and have the foresight to refer to other professionals.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organisation rule-based decision-making process. PS2. Take decisions with a systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognise problems, and search for solutions. PS8. Choose the best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day-to-day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

The equipment required includes: livestock handling facility, liquid nitrogen flask, fridge, handwashing facility, cleaning facility, toolbox or carrying basket. The tools required: AI pipette/gun/rod, warm water flask, scissors or straw cutter, straw-grabbing forceps, heat detection aides (where needed). The consumables required include: semen straws/sachets – coded and labelled, liquid nitrogen, long rectal gloves, short latex gloves, plastic sheaths, plastic sleeves (optional but recommended for optimum hygiene), liquid soap, paper towelling, record keeping book and pen.

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

Dilemmas associated with the job of an Artificial Insemination Technologist include long working hours, exposure to chemical (liquid nitrogen dangers), physical (injuries from cows, slipping, and exposure to dirt) and biological (zoonotic diseases) hazards, handling of situations where the technologist is requested to AI sick or malnourished animals, time pressure to complete tasks, working in extreme weather such as heat, cold and rain, team work requiring reliance on heat detectors, pressure to perform from supervisors and farmers.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include carrying out risk assessment and implementing appropriate control measures, ensuring good time management and planning, participating in workplace safety sensitization and awareness (especially with regard to liquid nitrogen, livestock handling and zoonoses), attending refresher trainings, adhering to company's safety and standard operating procedures at all times, paying attention to detail, refusing to AI sick or malnourished animals, training for heat detectors staff, consulting extensively within and outside one's department/team on safety and other issues.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include working in cold, hot and wet conditions, working in danger behind cows, stand/walk for long hours, inflexible hours, areas that are noisy and dirty, areas with limited lighting and ventilation. Working with dangerous product (liquid nitrogen)

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Parties involved/interacting with the job holder who are internal to the organisation include farmer, farm manager, animal breeders, supervisors, subordinates (milkmen, heat detectors), and other employees.

8.2 External/Outside the Organisation

Parties involved/interacting with the job holder who are external include government regulators, animal breeders, trainers, suppliers of equipment/tools/consumables, veterinarian surgeons, animal health technologist, occupational health and safety associations, Academia etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Strong arms with stamina to work gently in horizontal position for long hours
- Be able to lift relatively heavy materials, tools and equipment;
- Be able to blindly use fingers, hands with ease to complete the assigned task (dexterity); this requires an in-depth knowledge of each species internal anatomy
- Extreme gentleness

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

These guidelines pertain specifically to the training in artificial insemination itself. All persons who carry out training shall be diploma holders who either complete the specialized practical training as part of their course. A holder a Diploma in Animal health must receive both theoretical and practical training in artificial insemination, for them to be regarded as a Livestock Artificial Inseminator.

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

A1.3 The pass mark for the practical component shall be 80% in order to ensure that qualified technologists have a high chance of attaining successful pregnancies in the field. The pass mark for the theoretical component of the assessment shall be 60%.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.LAI.01		
ZQF Level	6	Version Number	01
Sector	Agriculture	Date of Approval	19 th May 2022
Sub Sector	Livestock	Date of Last Review	N/A
Occupation	Livestock Artificial Inseminator	Date of Next Review	May 2027

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